

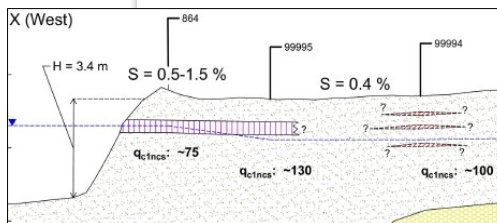
AI IN GEOTECHNICAL ENGINEERING

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J. Neils Thompson Centennial Teaching Fellow

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Janet S. Cockrell Centennial Chair in Engineering

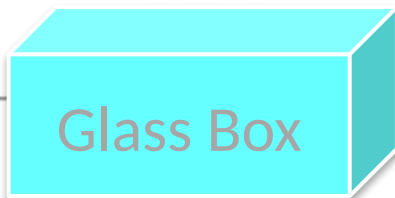
Department of Civil, Architectural, and Environmental Eng.
The University of Texas at Austin





Artificial Intelligence

Black Box Model

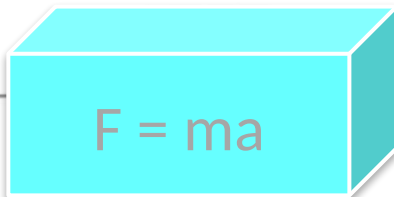


Lateral Spread?
Yes (75%)



Law of Physics

Black Box of Nature



Objectives

- XXXX

What else to use as intro to
AI/ML in geotech

Objectives

- Understand the data needs required for AI and Machine Learning (ML) in geotechnical engineering
- Gain a basic knowledge of a range of AI and ML methods and their application to geotechnical engineering
- Provide hands-on experience with training and interpreting AI and ML models
- Learn about emerging AI topics and their application to geotechnical engineering

Modules

- Module 1: Exploratory data analysis and feature engineering
- Module 2: Tree-based classification and regression models
- Module 3: Explainable AI (XAI) and Glass Box models
- Module 4: **Split this over two slides**
- Module 5: Introduction to Neural Networks (NN)
- Module 6: Application of Neural Networks
- Module 7: Recurrent NN for time series
- Module 8: Large Language Models (LLM)
- Module 9: Convolutional NN
- Module 10: Emerging topics in AI for geotechnical engineering

EXPLORATORY DATA ANALYSIS AND FEATURE ENGINEERING

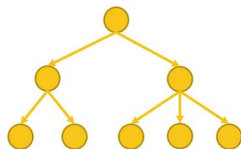


Exploratory Data Analysis (EDA)

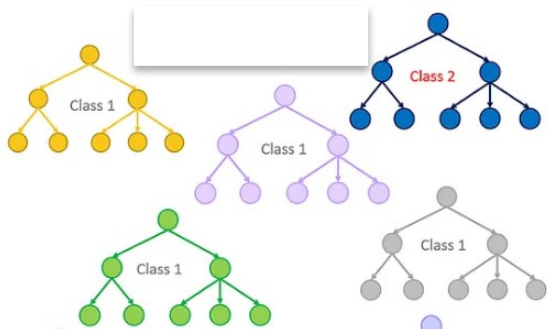
- **Goal:** better understand your data before using it to train an AI/ML model
 - Evaluate distribution of each of the variables
 - Identify errors, outliers, etc.
 - Identify trends and relationship between variables

AI and ML methods

Decision Tree

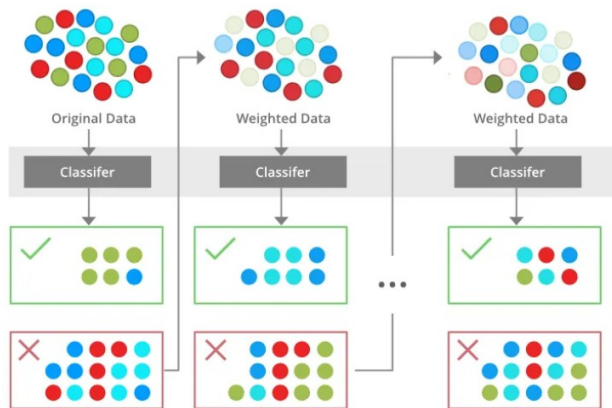


Random Forest

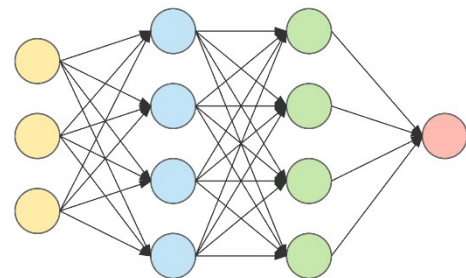


<https://tinyurl.com/treemethods>

XGBoost

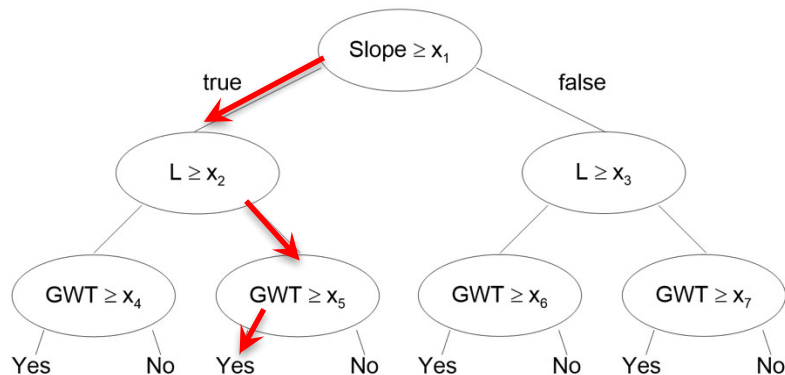


ANN



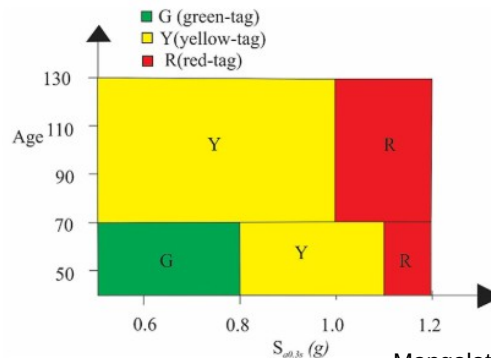
input layer hidden layer 1 hidden layer 2 output layer

Decision Tree



Yes

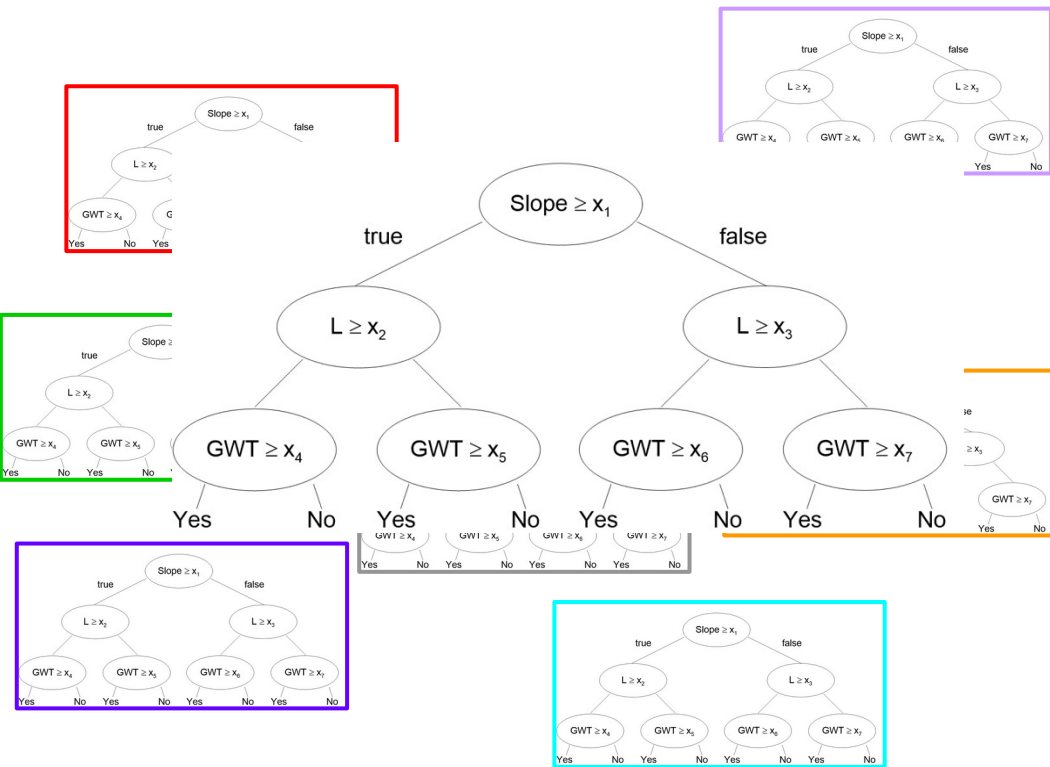
- Advantages
 - Simple and explainable



Mangalathu et al. (2020) EQS

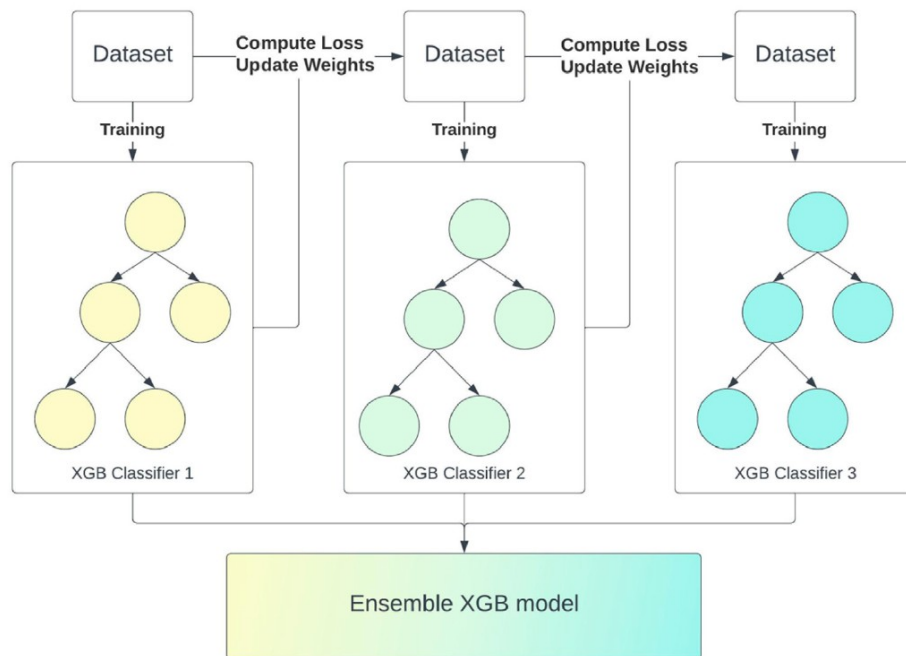
- Disadvantages
 - Overfitting/non-generalizable
 - Instability
 - Over-simplified

Random Forest



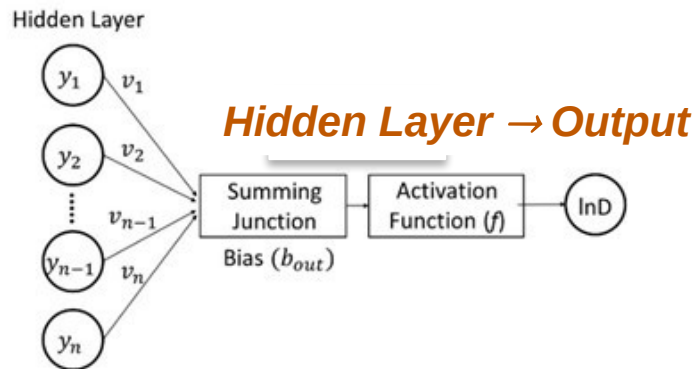
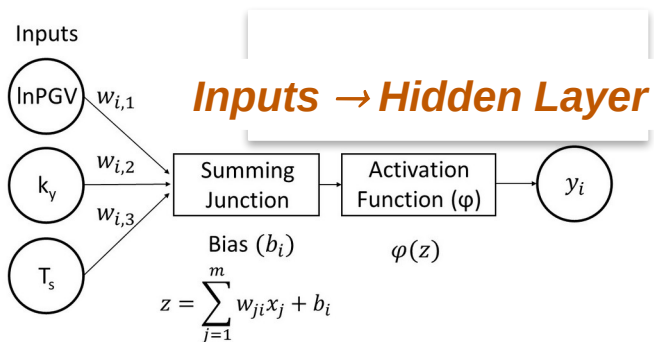
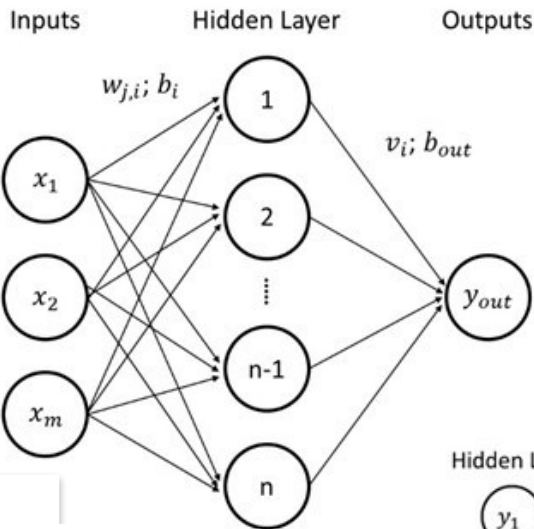
- Advantages
 - Increased accuracy
 - Less prone to overfitting
- Disadvantages
 - Increased complexity
 - Less explainable

XGBoost

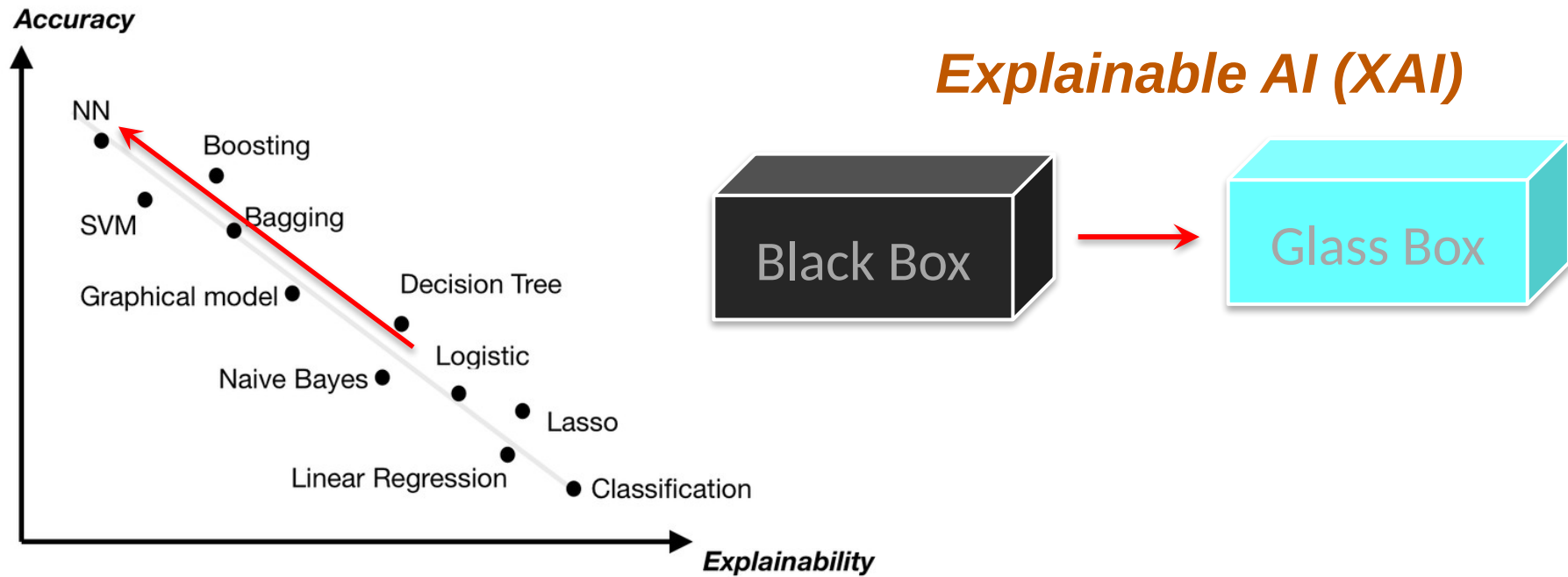


- Advantages
 - Increased accuracy
 - Less prone to overfitting
- Disadvantages
 - Increased complexity
 - Less explainable

Artificial Neural Network (ANN)



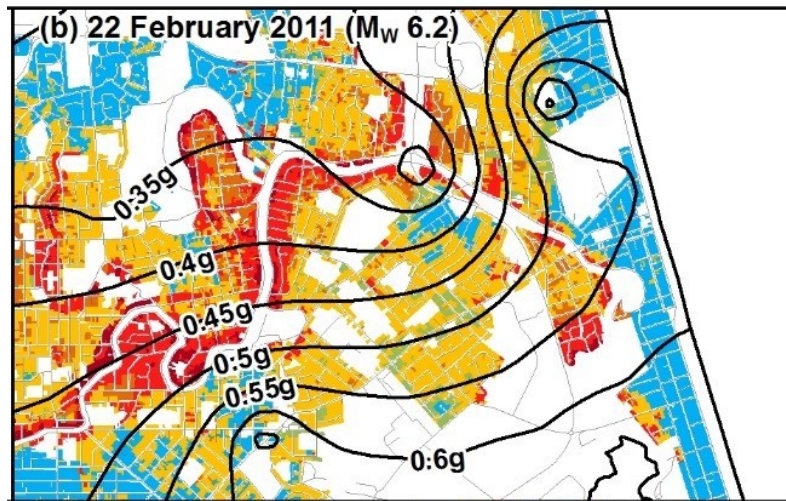
Accuracy vs. Explainability



Outline

- AI and ML methods
 - Tree-based methods, artificial neural networks (ANN)
 - Interpretability and explainability
- Liquefaction-induced lateral spreading
 - Prediction model developed from field observations
 - Evolution of model towards explainability and incorporating knowledge
- Seismic slope displacement
 - Model developed from finite element simulations
 - Traditional regression vs. AI / ML methods
- Data and computational needs for AI / ML

2011 Christchurch Earthquake



- No observed ground cracking or ejected liquefied material
- Minor ground cracking but no observed ejected liquefied material
- No lateral spreading but minor to moderate quantities of ejected material
- No lateral spreading but large quantities of ejected material
- Moderate to major lateral spreading; ejected material often observed
- Severe lateral spreading; ejected material often observed

Liquefaction Effects

- 60,000 of 150,000 single family homes affected; 8,000 homes demolished
- No rebuilding in red-zone along river
- Half of \$30 billion NZD losses associated with liquefaction



Observed Lateral Spreading

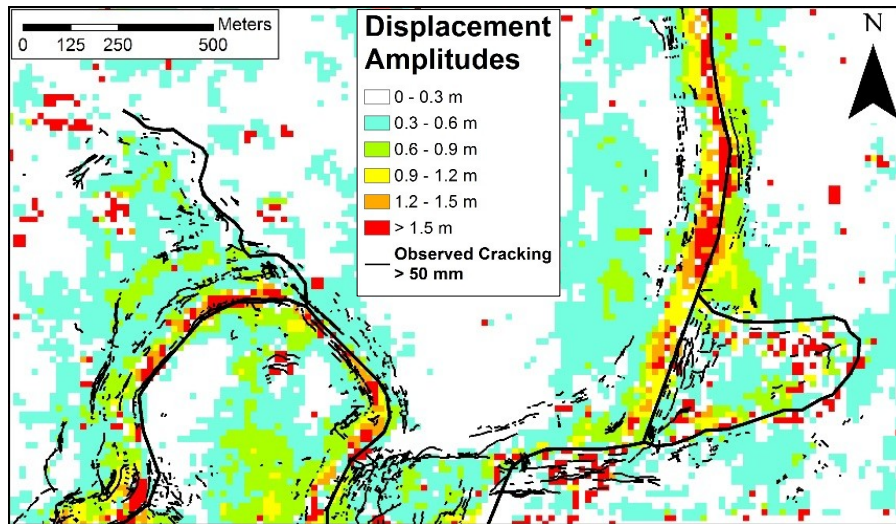
Qualitative Observations



Observed Liquefaction Related Land Damage

- | | |
|---|---|
| ■ No observed ground cracking or ejected liquefied material | ■ No lateral spreading but large quantities of ejected material |
| ■ Minor ground cracking but no observed ejected liquefied material | ■ Moderate to major lateral spreading; ejected material often observed |
| ■ No lateral spreading but minor to moderate quantities of ejected material | ■ Severe lateral spreading; ejected material often observed |

Displacements from Satellite Imagery



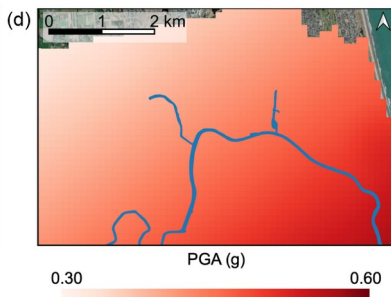
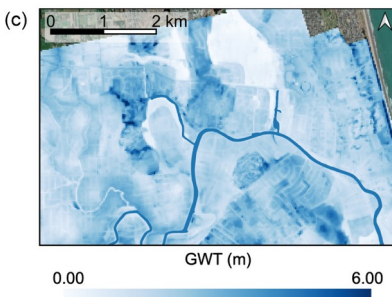
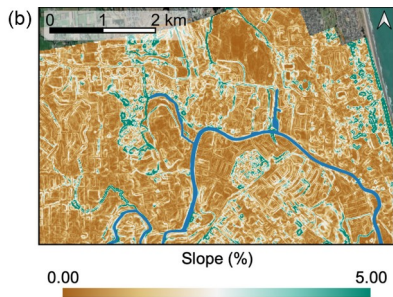
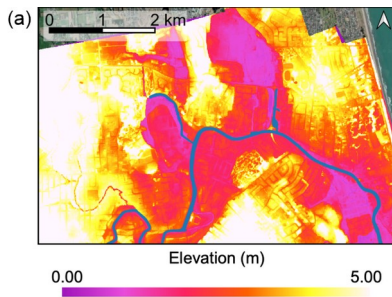
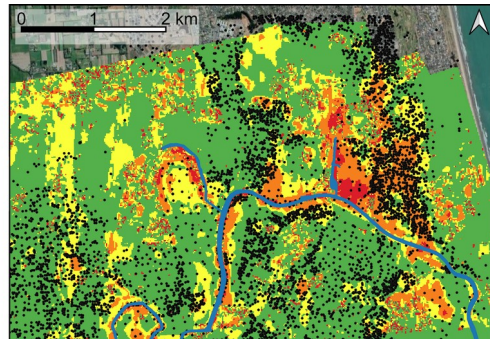
Rathje et al. (2017) *EQ Spectra*, <https://doi.org/10.1193/080816eqs127m>

Predictive Lateral Spread Model

- **Goal:** Predict occurrence of lateral spreading and associated displacement using machine learning
- **Given:** Ground motion, geomorphic, topographic, and subsurface data

Displacement classes (m)

- None (< 0.30)
- 0.30 - 0.50
- 0.50 - 1.00
- > 1.00
- CPT locations



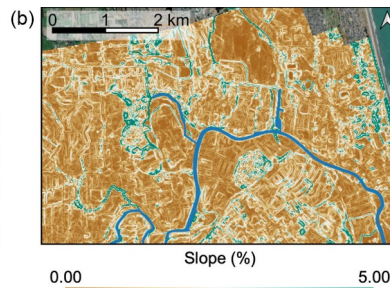
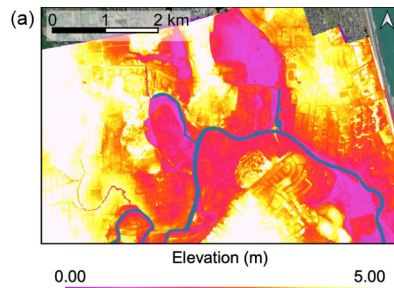
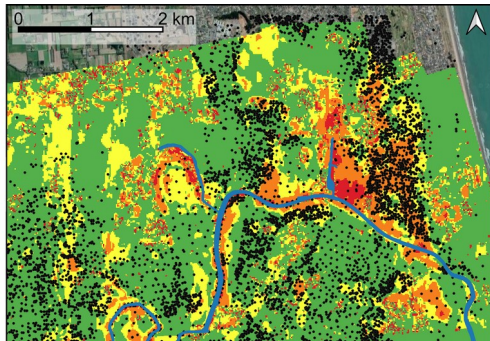
Input Data

Displacements

Displacement classes (m)

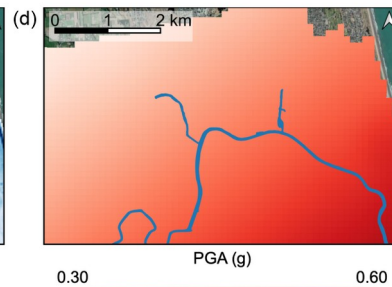
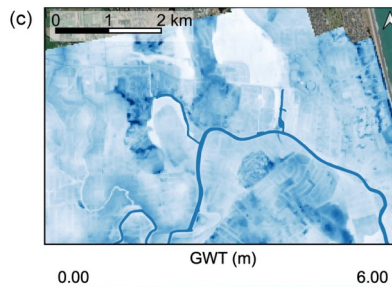
- None (< 0.30)
- 0.30 - 0.50
- 0.50 - 1.00
- > 1.00

- CPT locations

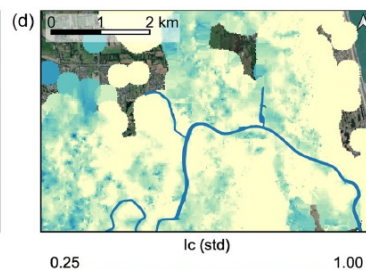
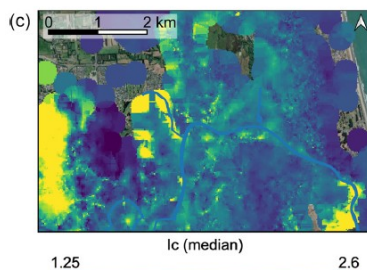
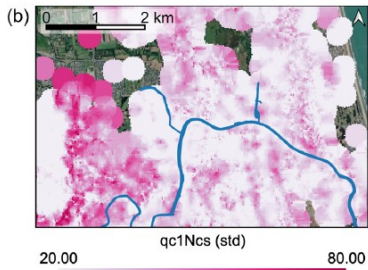
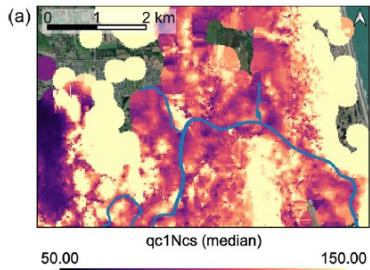


Elevation, Slope

GWD, PGA, Dist to River (L)



CPT (for 4 m below GWT)



Training Data

- ~7300 pts where displacement and CPT data are both available
 - Inputs (PGA, L to river, Slope, Elev, GWD) from regional data
 - Displacements from optical image correlation (Rathje et al. 2017)
 - CPT data from NZ Geotechnical Database (NZGD)
- Random forest classification with K-fold cross validation (80%-20% train-test split)

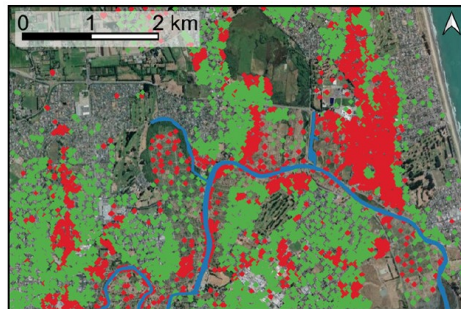
Occurrence

Class 0	No Lateral Spread	4236 pts
Class 1	Yes Lateral Spread	3055 pts

Displacement

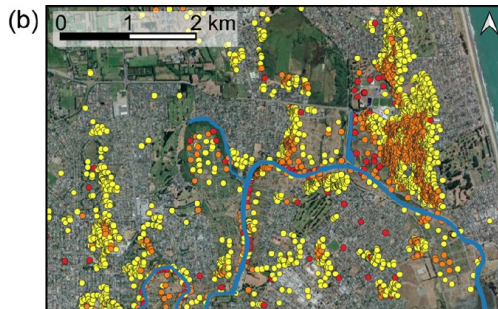
Class 0	0.3–0.5 m	1799 pts
Class 1	0.5–1.0 m	1138 pts
Class 2	>1.0 m	118 pts

7291 pts



Lateral Spreading: ♦ YES ♦ NO

3055 pts



Displacement Classes (m):
 ♦ 0.30 - 0.50 ♦ > 1.00
 ♦ 0.50 - 1.00

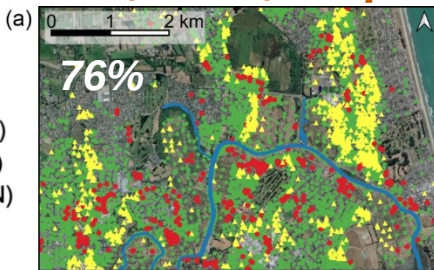
Results: Lateral Spread Occurrence

L, GWT, Slope

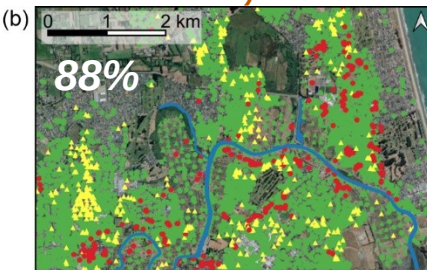
+ PGA, Elev

+ CPT

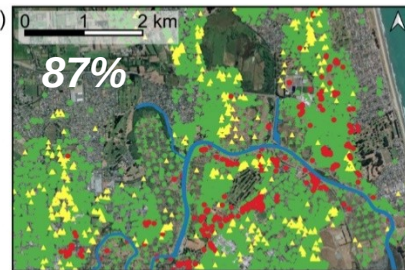
- ▲ True Positive (TP)
- True Negative (TN)
- False Positive (FP)
- ▲ False Negative (FN)



Model 0



Model 3

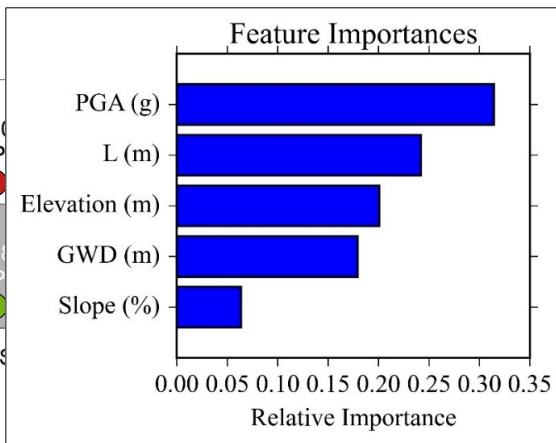


Model 5

Overall Accuracy
76-88%

(a) Model 0

Observed	NO	0.90 (TN) ▲	0.10 (FP) ●
	YES	0.42 (FN) ▲	0.58 (TP) ●
	Predicted	NO	YES



Model 5

Observed	NO	0.93 (TN) ▲	0.07 (FP) ●
	YES	0.22 (FN) ▲	0.78 (TP) ●
	Predicted	NO	YES

Results: Lateral Spread Displacement

Observations

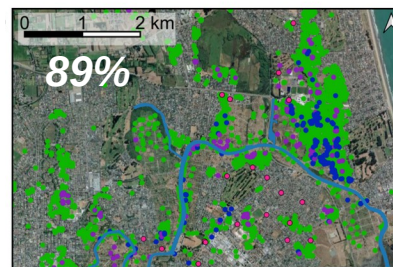
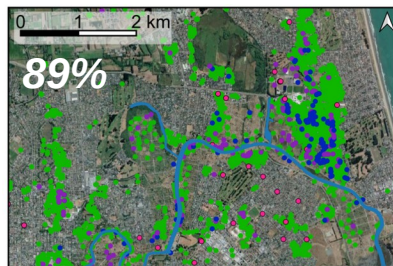
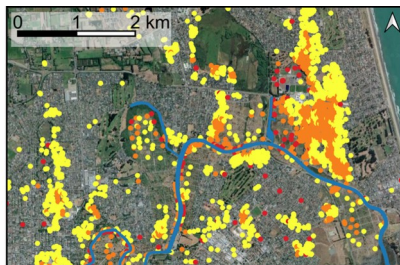
L, GWT, S, PGA, EI

+ CPT

**Overall
Accuracy
89%**

Observed displacement
classes (m):

- 0.30 - 0.50
- 0.50 - 1.00
- > 1.00



● PC - OC = -2 ● PC - OC = -1 ● PC - OC = 0 ● PC - OC = +1
 PC: Predicted Class; OC: Observed Class

Observed	0.3 - 0.5 m	0.92 ●	0.08 ●	0.00
	0.5 - 1.0 m	0.11 ●	0.89 ●	0.00
	> 1.0 m	0.25 ●	0.33 ●	0.42 ●
		0.3 - 0.5 m	0.5 - 1.0 m	> 1.0 m
		Predicted		

Observed	0.3 - 0.5 m	0.91 ●	0.09 ●	0.00
	0.5 - 1.0 m	0.10 ●	0.90 ●	0.00
	> 1.0 m	0.19 ●	0.32 ●	0.49 ●
		0.3 - 0.5 m	0.5 - 1.0 m	> 1.0 m
		Predicted		

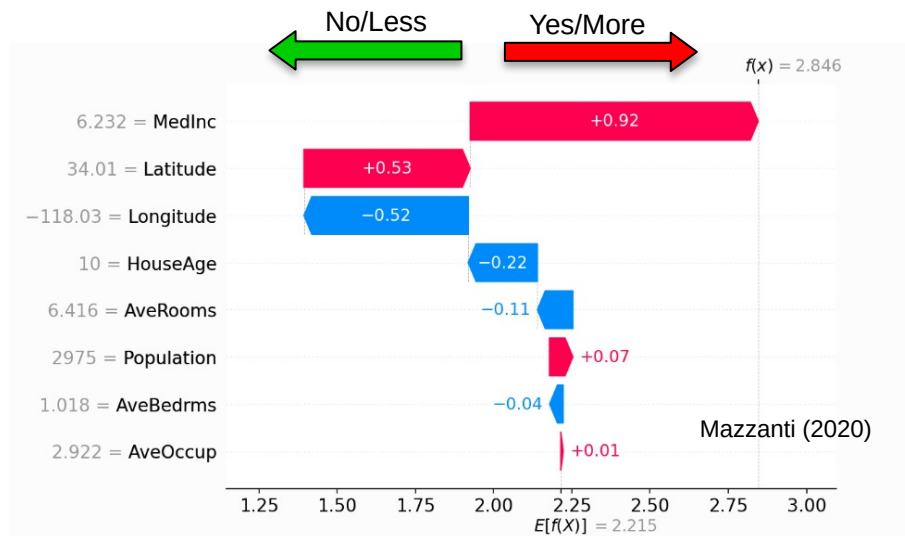
1799 pts

1138 pts

118 pts

Explainability of ML models

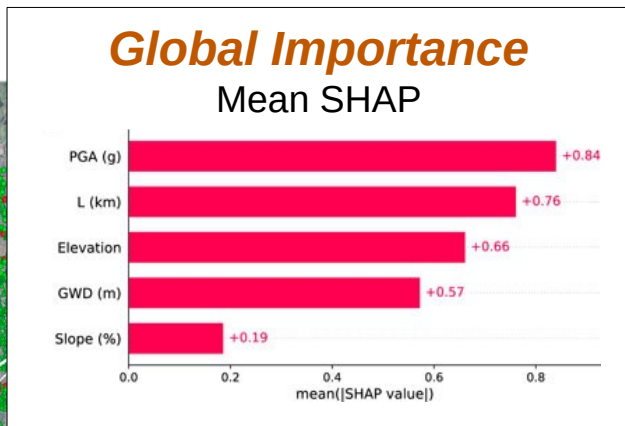
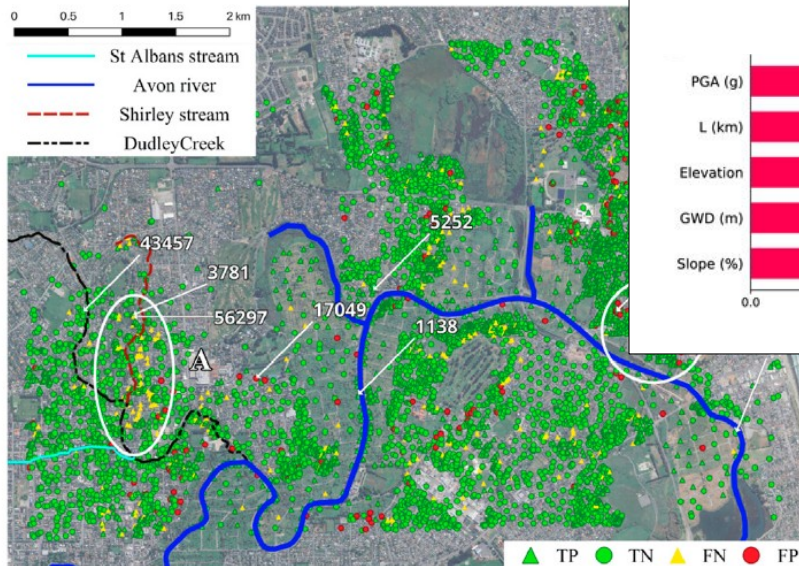
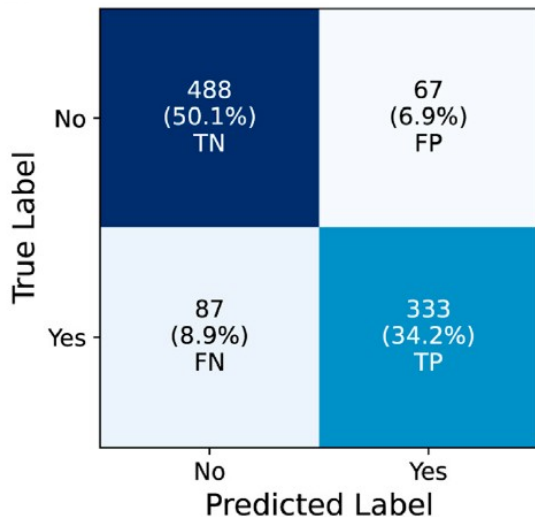
- Utilize **SH**apley **A**dditive ex**P**lanations (SHAP) values
 - Developed from game theory
 - Estimates the contribution of each feature to a prediction
 - Local explanation (i.e. individual prediction) vs. global explanation



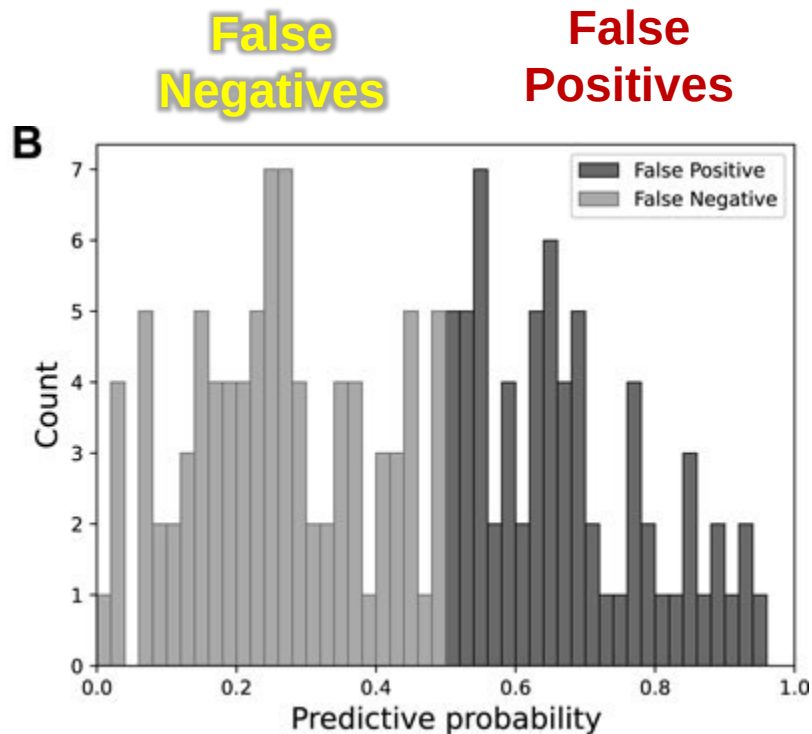
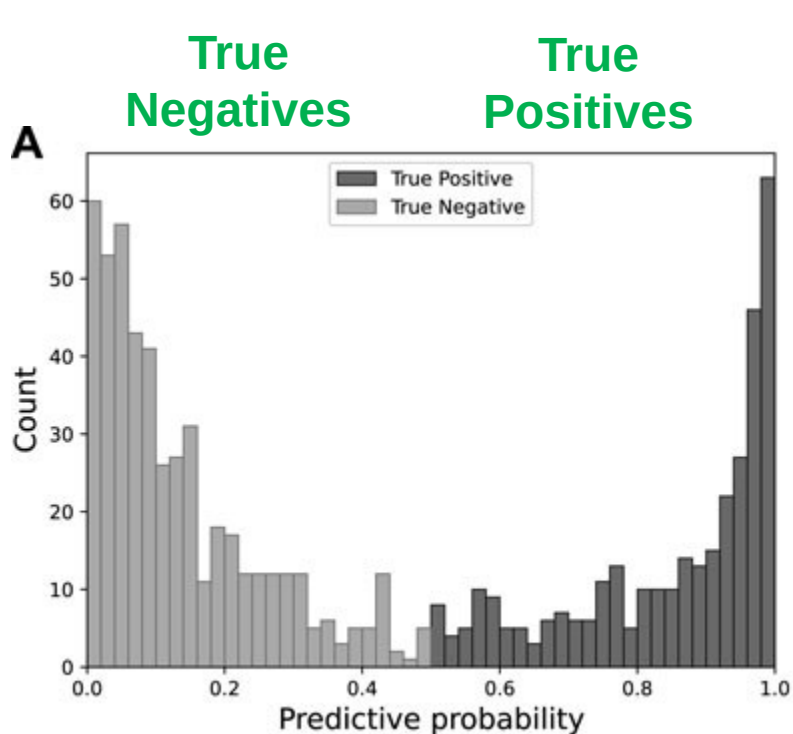
Waterfall Plot
 SHAP for each feature for a
 individual prediction

XGBoost model and SHAP

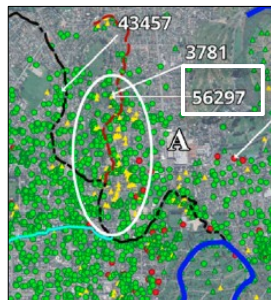
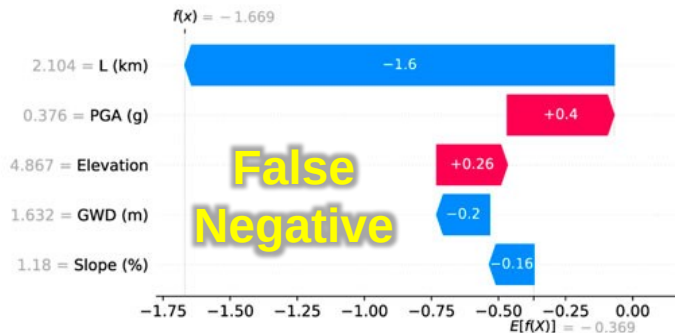
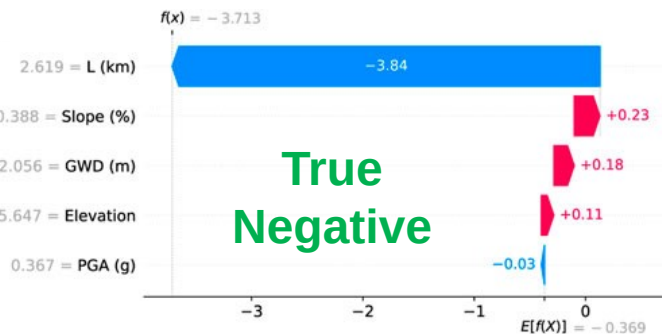
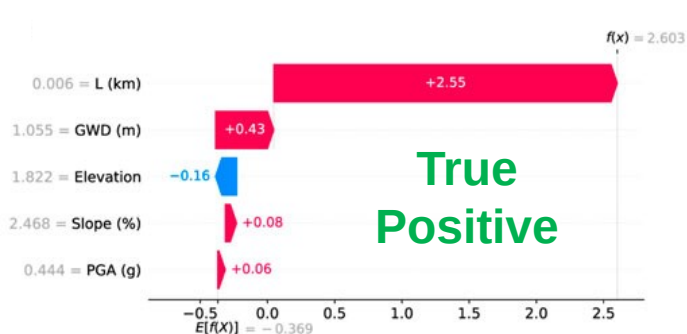
Overall Accuracy: 84%



Predictive probabilities

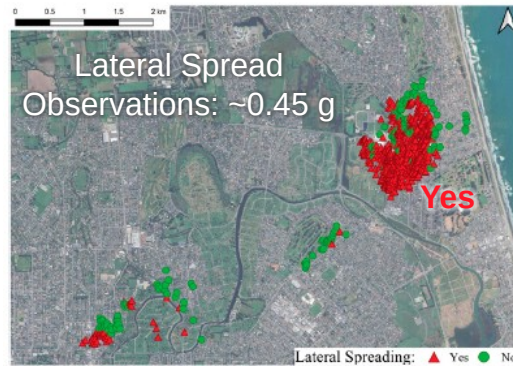
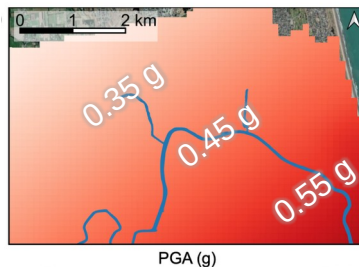
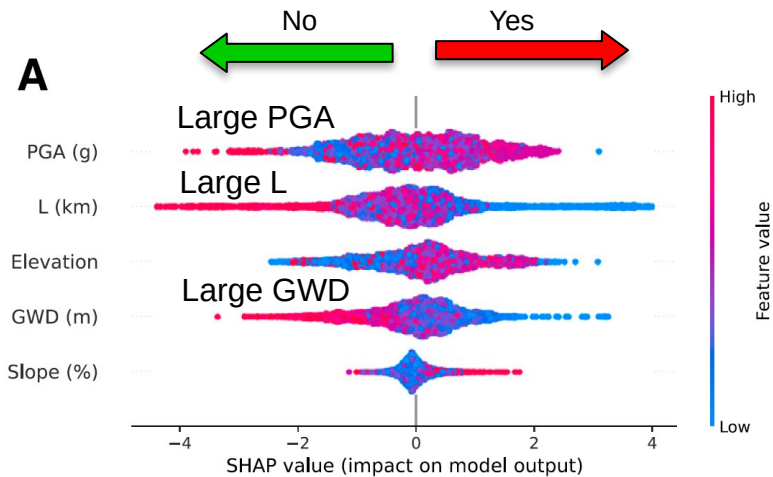


SHAP: local waterfall plots

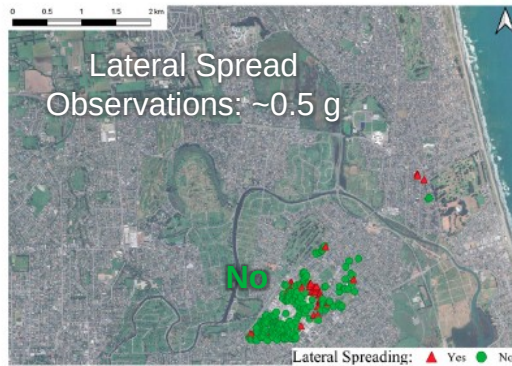
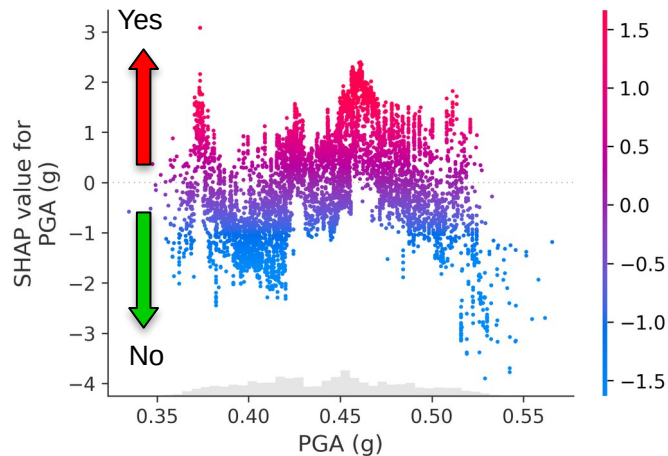


SHAP: beeswarm plots

SHAP for all predictions



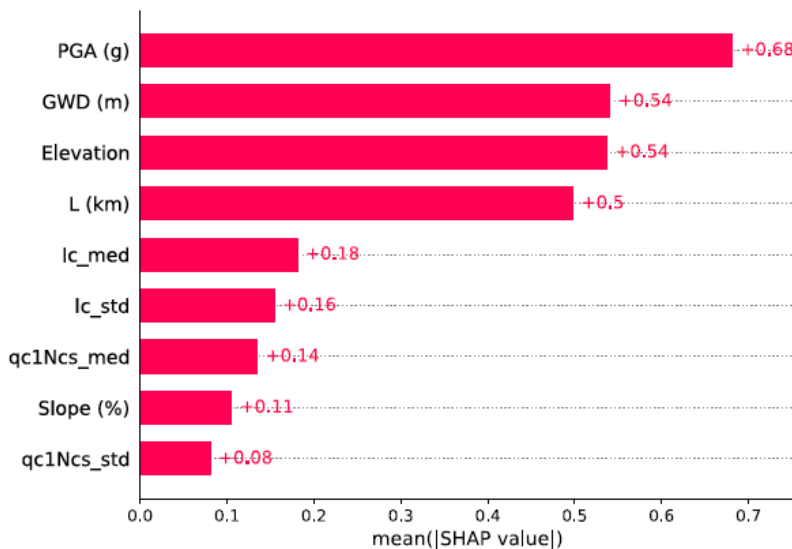
SHAP vs. PGA



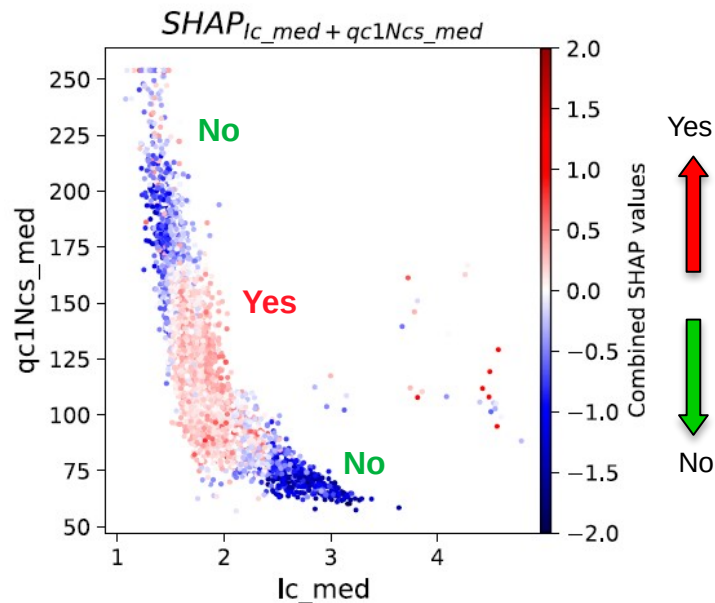
Incorporating CPT features

Overall Accuracy: 85% (vs. 84% for no CPT)

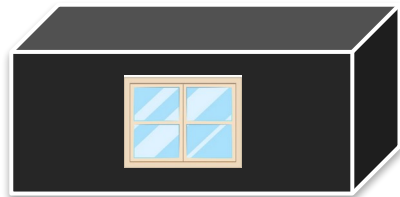
Global Importance



Feature Relation + SHAP



Beyond Explainability...

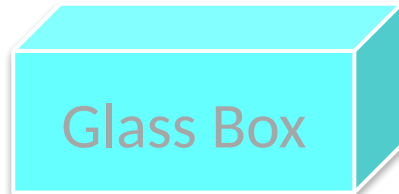
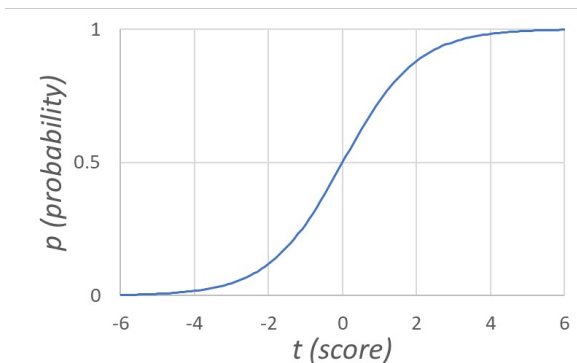


- SHAP values provide post-hoc explainability
 - Model may learn “bad trends” based on characteristics of the dataset
 - Influenced by trends in field observations, as well as missing input features

Can we incorporate domain knowledge into ML model??

- Explainable Boosting Machine (EBM)
 - Inspired by logistic regression and generalized additive models (GAM)

$$p = \frac{1}{1 + e^{-t}}$$



Explainable Boosting Machines (EBM)

$$p = \frac{1}{1 + e^{-t}}$$

- Logistic regression using input features x_1 to x_n

$$t(\mathbf{x}) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots$$

- Generalized additive model (GAM) using x_1 to x_n

$$t(\mathbf{x}) = \beta_0 + f_1(x_1) + f_2(x_2) + \dots$$

- Explainable Boosting Machine (EBM) using x_1 to x_n

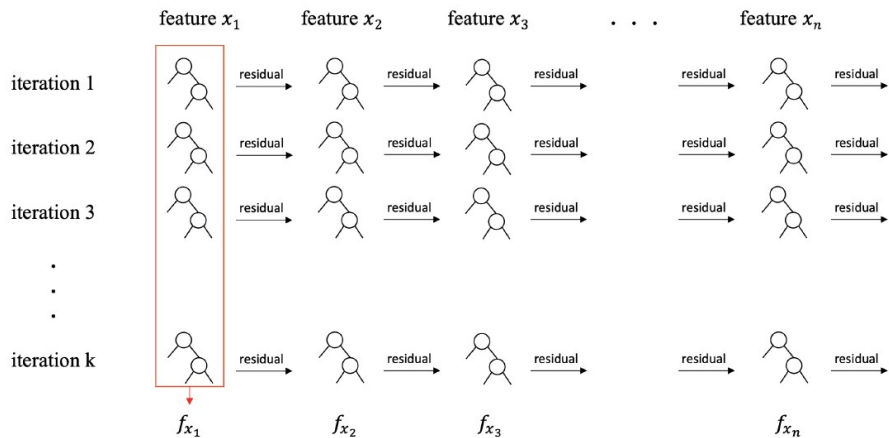
$$t(\mathbf{x}) = \beta_0 + f_1(x_1) + f_2(x_2) + \dots + f_{1,2}(x_1, x_2) + \dots$$

Score Functions

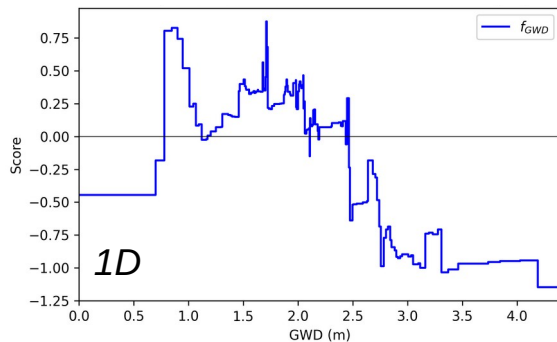
Explainable Boosting Machines (EBM)

Score Functions

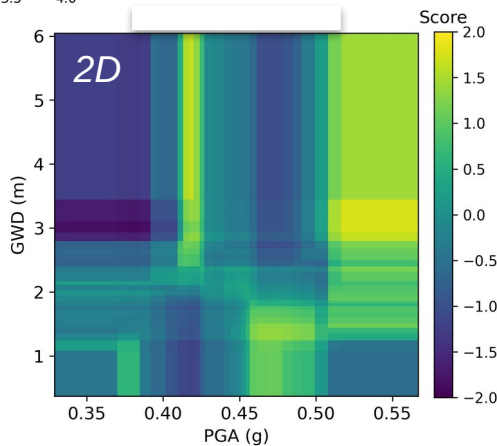
Training Process



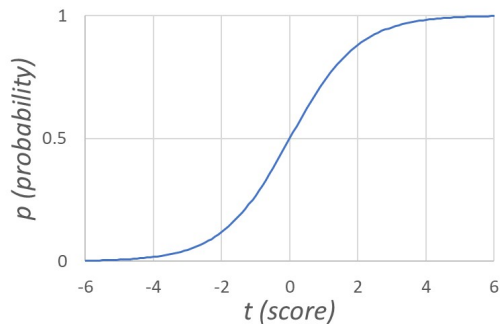
One iteration for 2D pair-wise interactions $f_{i,j}(x_i, x_j)$



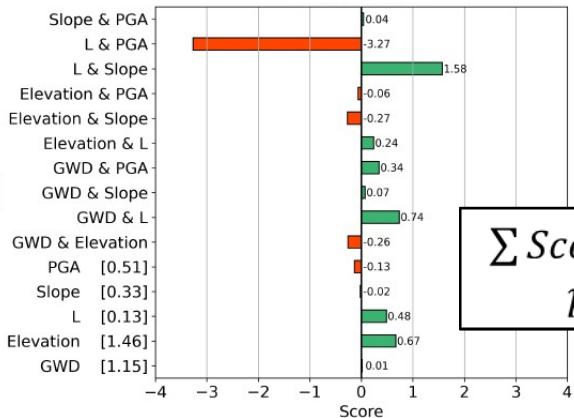
Non-parametric



Model Usage and Physical Behaviors



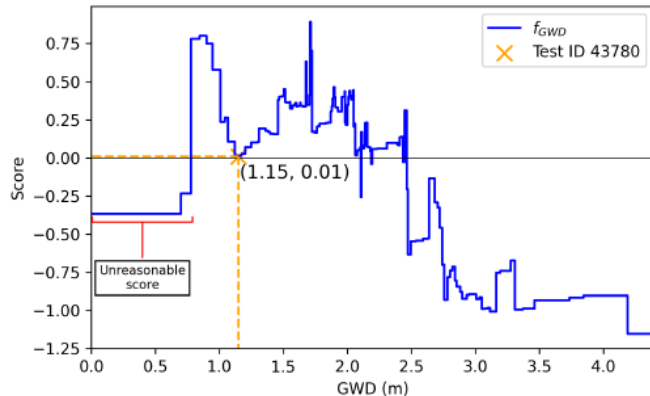
No ← → Yes



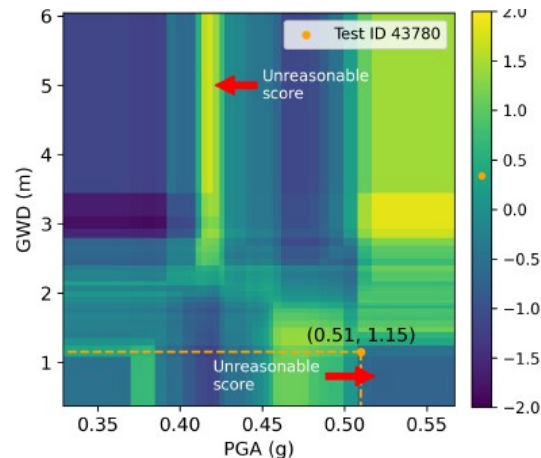
$$\sum Scores_i = -0.39$$

$$p = 0.41$$

Score Function: GWD



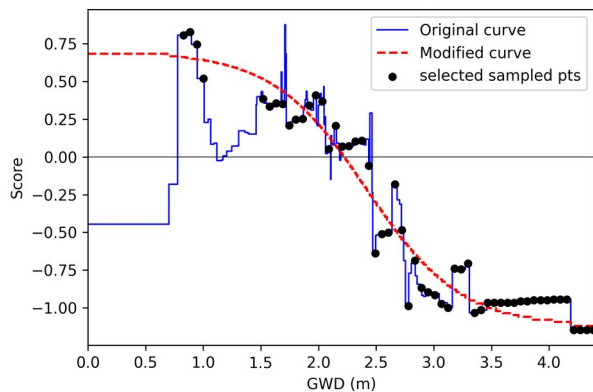
Score Function: GWD + PGA



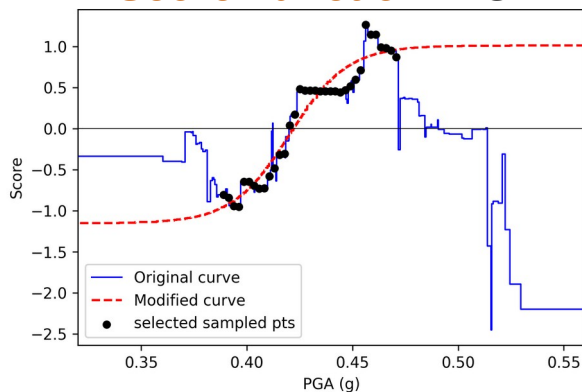
Adjusting 1D Score Functions

Guided by domain knowledge

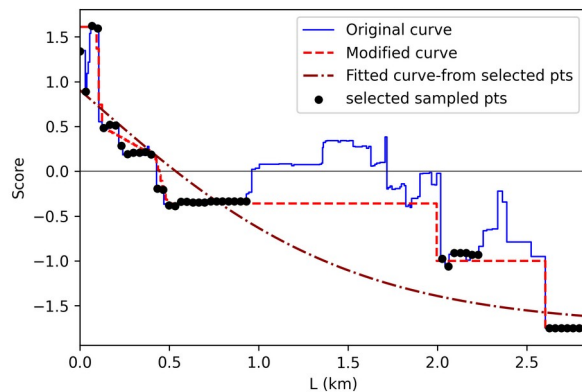
Score Function: GWD



Score Function: PGA



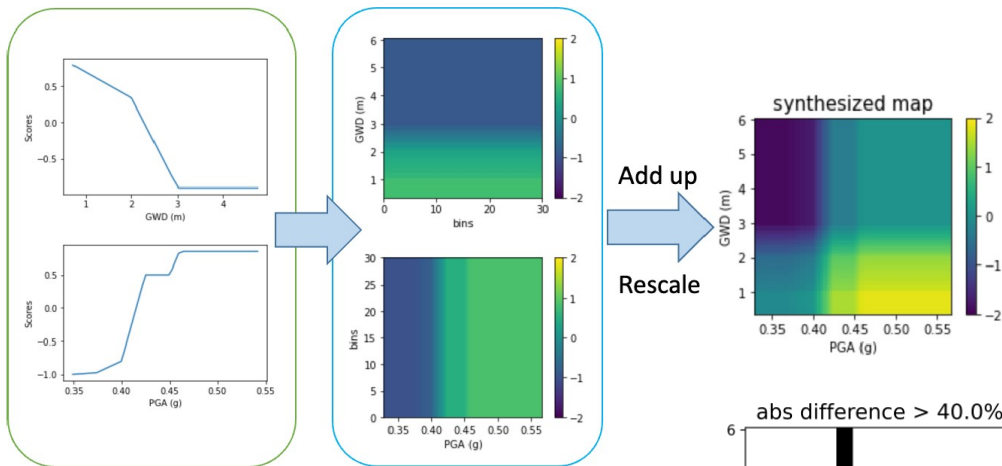
Score Function: L



Adjusting 2D Score Functions

Synthetic 2D Score Function

Based on 1D score functions



1D Adjustments

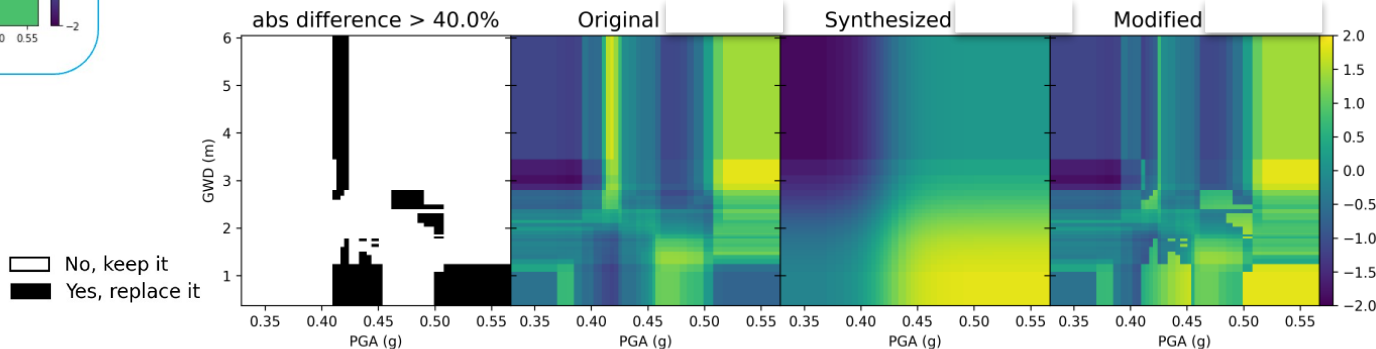
- GWD, PGA, L

2D Adjustments

- GWD + PGA
- L + PGA
- GWD + L

Modified 2D Score Function

Compare original and synthetic score functions

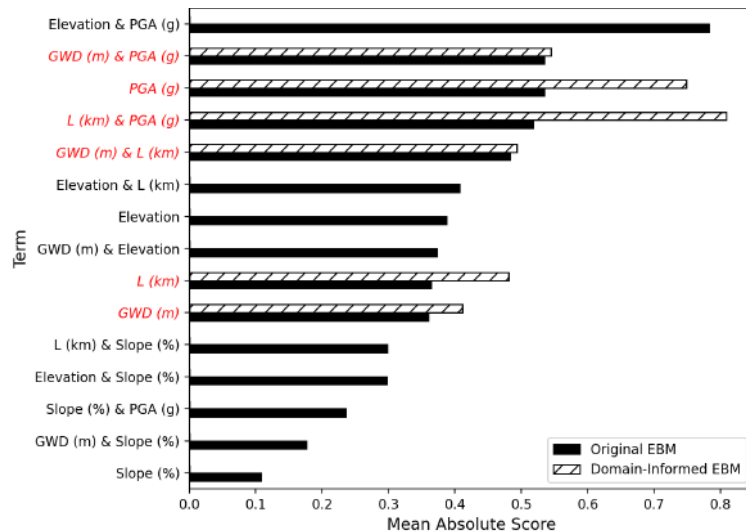


Impact on Lateral Spread Predictions

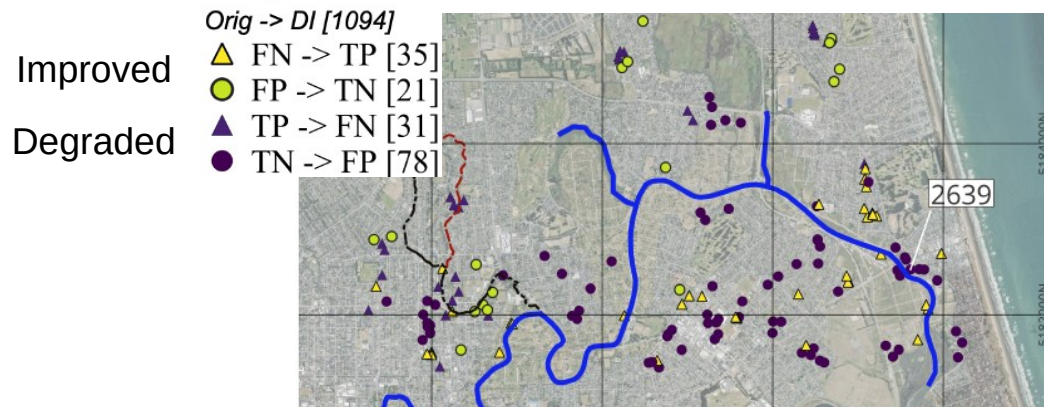
Accuracy Metrics

Dataset	Original EBM		Domain-Informed EBM	
	Validation	Testing	Validation	Testing
Accuracy	0.80	0.80	0.76	0.75
Precision	0.80	0.82	0.72	0.72
Recall	0.71	0.70	0.70	0.71
F1-score	0.76	0.75	0.71	0.71
AUC	0.87	0.88	0.80	0.80

Global Explanations



Changes in Predictions

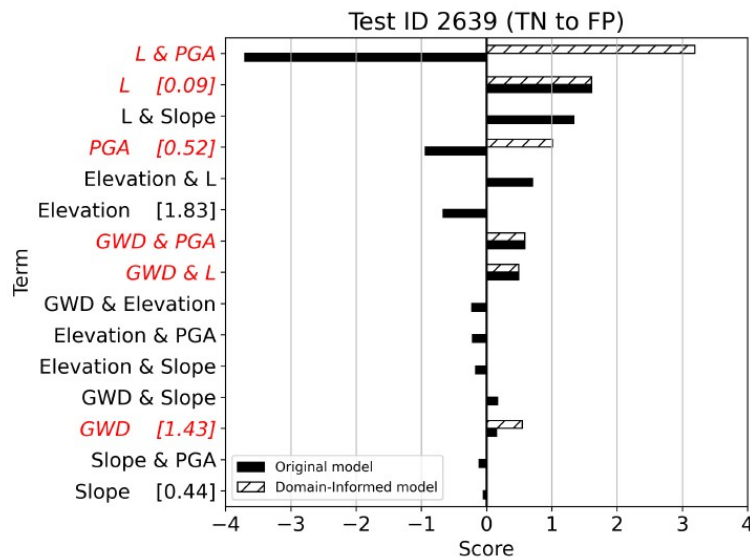


Impact on Lateral Spread Predictions

Accuracy Metrics

Dataset	Original EBM		Domain-Informed EBM	
	Validation	Testing	Validation	Testing
Accuracy	0.80	0.80	0.76	0.75
Precision	0.80	0.82	0.72	0.72
Recall	0.71	0.70	0.70	0.71
F1-score	0.76	0.75	0.71	0.71
AUC	0.87	0.88	0.80	0.80

Local Explanation

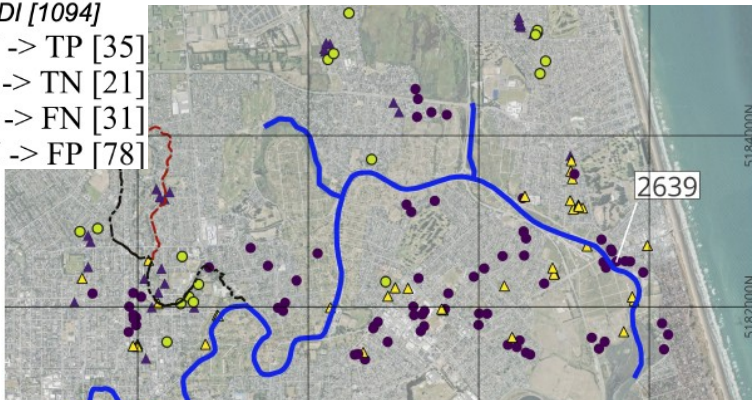


Changes in Predictions

Improved
Degraded

Orig -> DI [1094]

- ▲ FN -> TP [35]
- FP -> TN [21]
- ▲ TP -> FN [31]
- TN -> FP [78]



Comments

- All ML models should address explainability and move towards glass box approaches
 - Analyze SHAP values to understand influence of input features on results
 - Particularly important for models based on field observations
 - Incorporating domain knowledge may make model less accurate, but more trustworthy
- ML models applied to liquefaction-induced lateral spreading
 - How best to incorporate subsurface information?
 - Generalizable models require spatially-dense global datasets that are balanced (Y/N) with consistent input features for each earthquake event

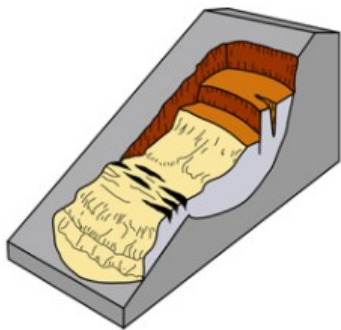
Outline

- AI and ML methods
 - Tree-based methods, artificial neural networks (ANN)
 - Interpretability and explainability
- Liquefaction-induced lateral spreading
 - Prediction model developed from field observations
 - Evolution of model towards explainability and incorporating knowledge
- Seismic slope displacement
 - Model developed from finite element simulations
 - Traditional regression vs. AI / ML methods
- Data and computational needs for AI /ML

***Domain knowledge
embedded***

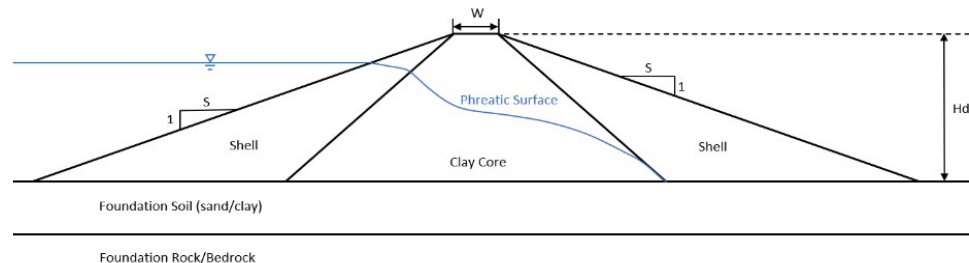
Seismic Performance of Earth Slopes

Earth Slope

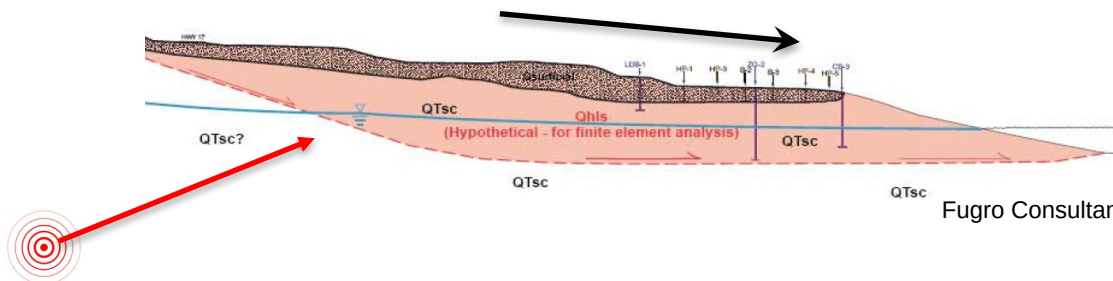


www.idahogeology.org/geologic-hazards/landslides

Earth Dam



$D =$ seismic movement

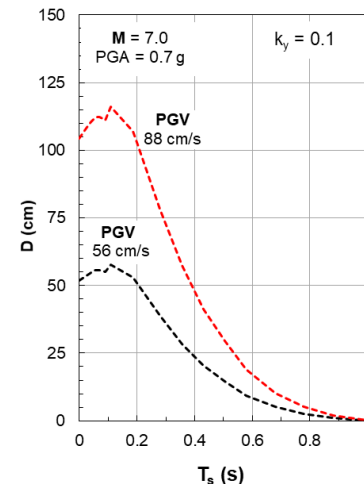
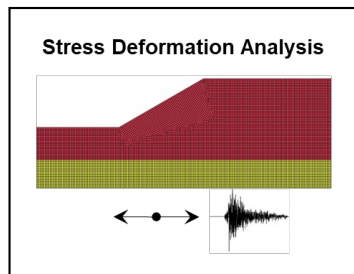
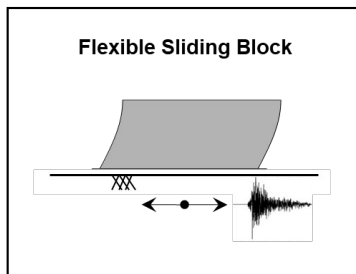
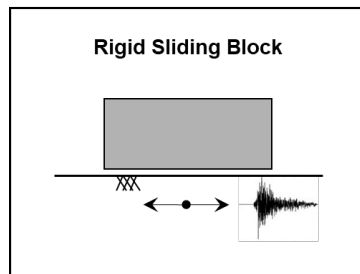
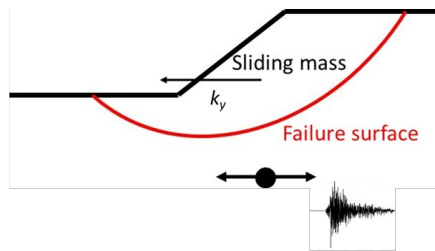


Fugro Consultants (2011, 2012)

Predicting Seismic Slope Movement

'Surrogate' models

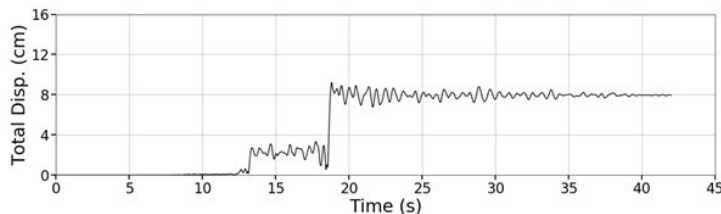
- Predict $D = f(x)$ (slope characteristics, ground shaking)
- Derived from a large suite of analyses



Rathje (2025) "Evolution of Approaches to Evaluate the Seismic Performance of Earth Slopes," <https://doi.org/10.1007/s40098-025-01181-y>

Stress Deformation Analysis

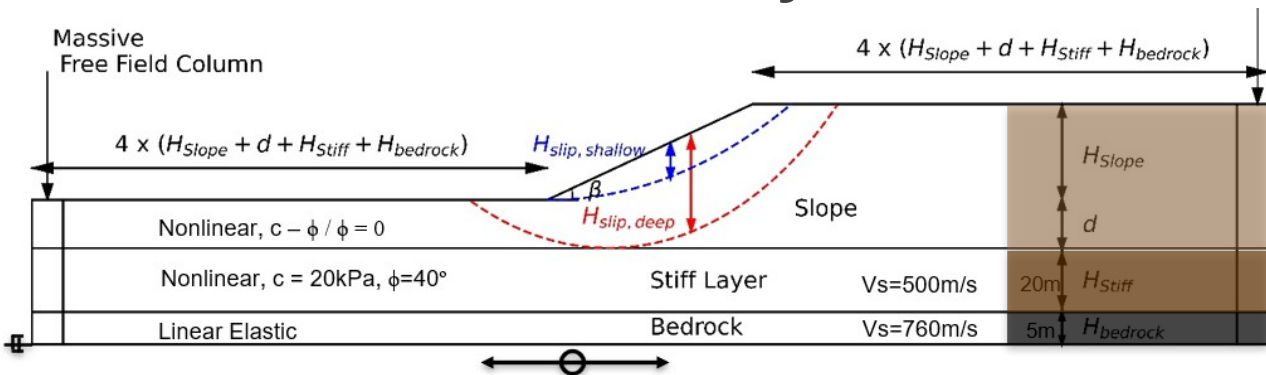
Finite Element / Difference (FE/FD) Analyses



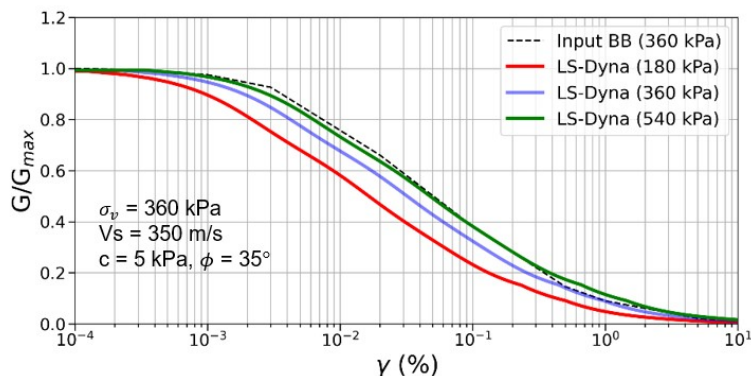
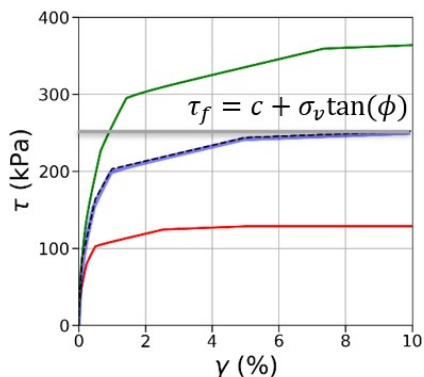
- No assumed slip surface
- Deformation does not need to be localized
- Nonlinear stress-strain response
- Accumulates movements before failure

Develop surrogate models for D computed by finite element simulations

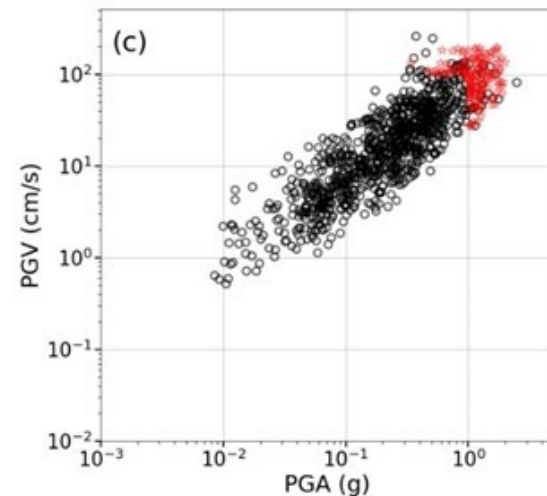
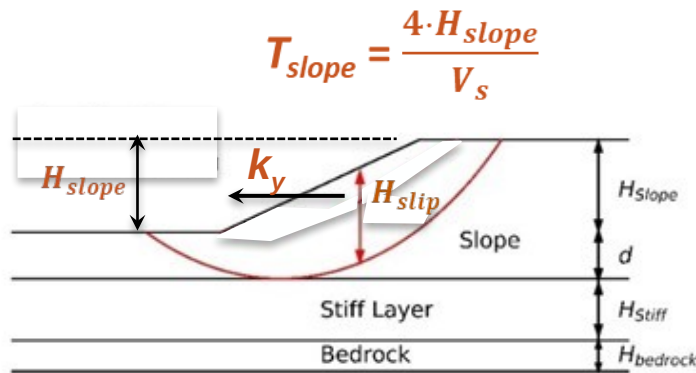
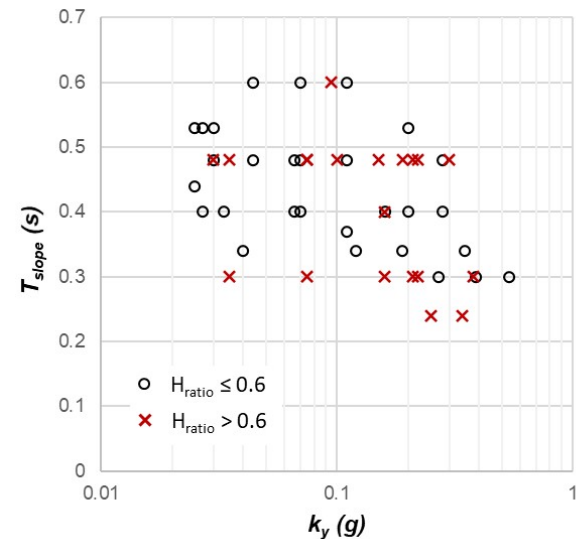
Finite Element Study



- $H_{\text{slope}} = 15, 30 \text{ m}$
- $d = (0, 0.5, 1.0) \times H_{\text{slope}}$
- $\beta = 20, 30, 40^\circ$
- Strength: $c - \phi$ or $\phi = 0$
- $V_s = 200 - 400 \text{ m/s}$
- Hyperbolic $\tau - \gamma$
- Analyzed using LS-DYNA



49 Slope Models x 1,000+ motions



Slope parameters: k_y : yield acceleration from pseudo-static analysis
 T_{slope} : flexibility of slope
 H_{ratio} : H_{slip}/H_{slope} , slip surface thickness relative to slope height

Check the physics...

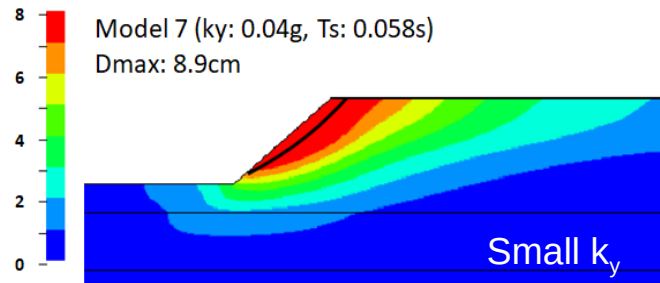
Shallow

(friction-
dominated)

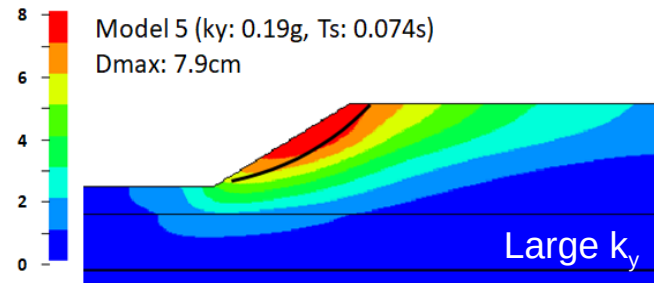
Resultant Displacement (cm)

—— Slip Surface from Pseudo-Static Analysis

Model 7 (k_y : 0.04g, T_s : 0.058s)
Dmax: 8.9cm



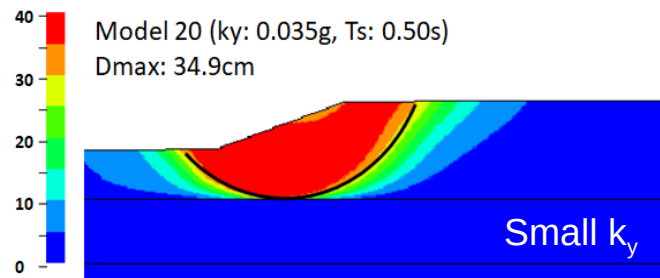
Model 5 (k_y : 0.19g, T_s : 0.074s)
Dmax: 7.9cm



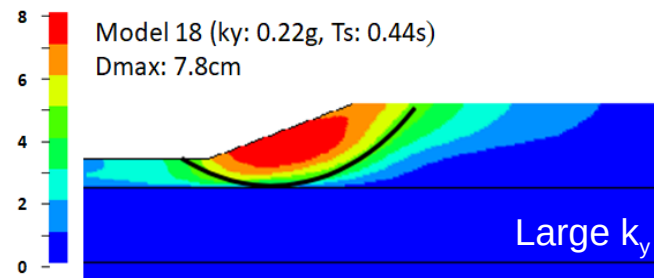
Deep

(cohesion-
dominated)

Model 20 (k_y : 0.035g, T_s : 0.50s)
Dmax: 34.9cm



Model 18 (k_y : 0.22g, T_s : 0.44s)
Dmax: 7.8cm



- Max. displacement (LS-DYNA) consistent with pseudostatic slip surface
- Distributed shear deformation below shallow slip surfaces due to nonlinear stress-strain response

Surrogate Model: Classical Regression

$$\ln(D) = a_0 + a_1 \cdot \ln(PGV)$$

For $H_{ratio} \leq 0.6$ (shallow):

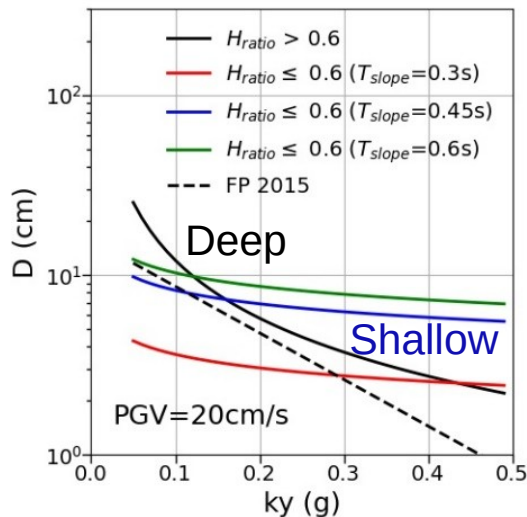
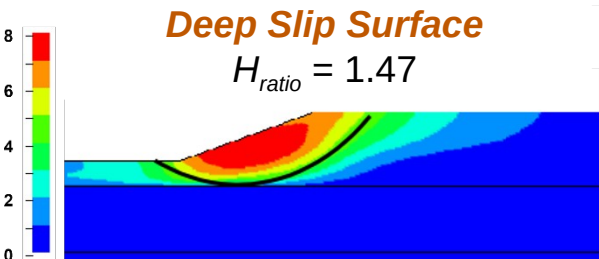
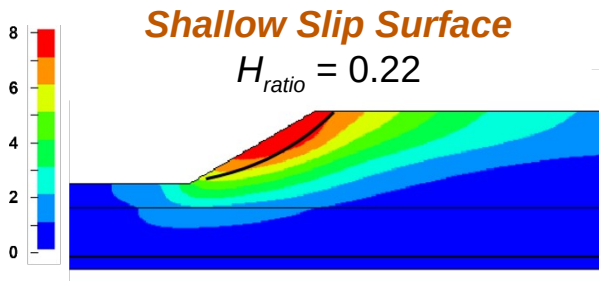
$$a_0 = \text{fxn}(T_{slope}, k_y)$$

$$a_1 = \text{fxn}(T_{slope})$$

For $H_{ratio} > 0.6$ (deep):

$$a_0 = \text{fxn}(k_y)$$

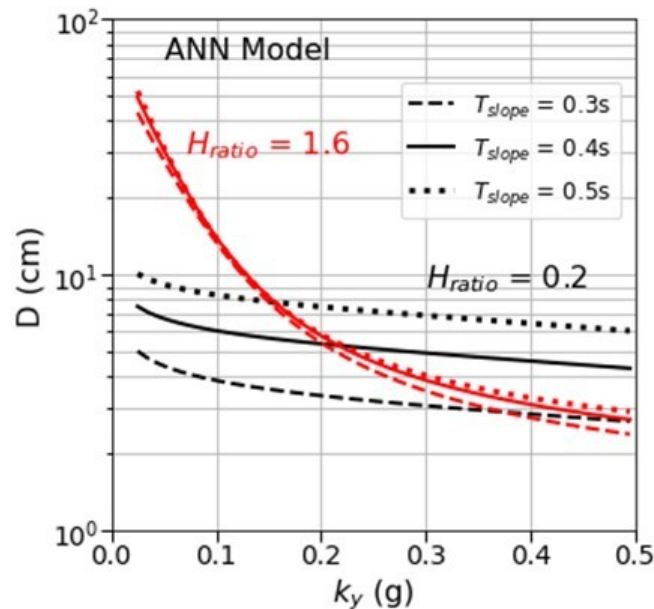
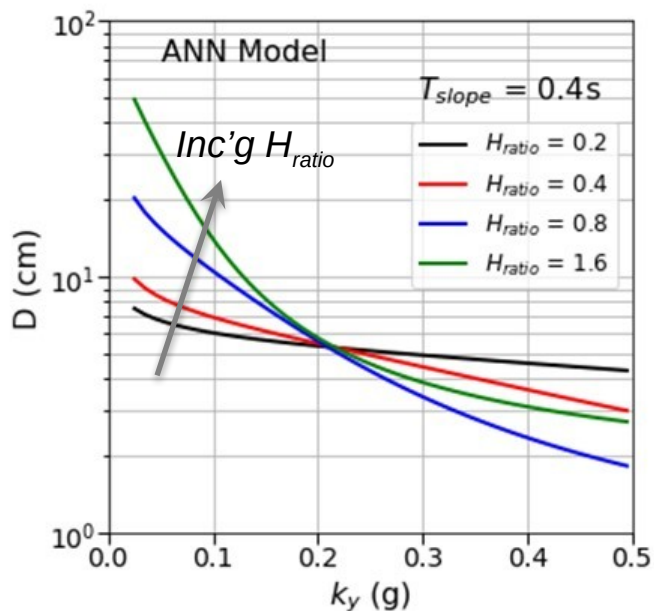
$$a_1 = \text{fxn}(k_y)$$



ANN Model

Cho et al. (2022), Soil Dyn and Earthquake Eng, <https://doi.org/10.1016/j.soildyn.2021.107024>

- k_y , T_{slope} , **and** H_{ratio} as input variables
- Continuous variation of D prediction with H_{ratio}
- Slope-to-slope variability reduced ~ 25%



Outline

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 - Traditional regression vs. AI / ML methods
- Data and computational needs for AI / ML



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Supporting Research in Natural Hazards through:

- A data repository for ***data publishing***
- Access to ***tools*** and ***computational resources*** for data analytics (AI/ML), simulation, etc.

Lowering the barrier of entry to computational research for the global natural hazards community



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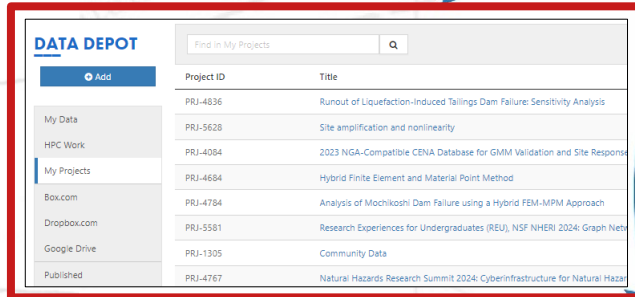
RICE

Florida Tech

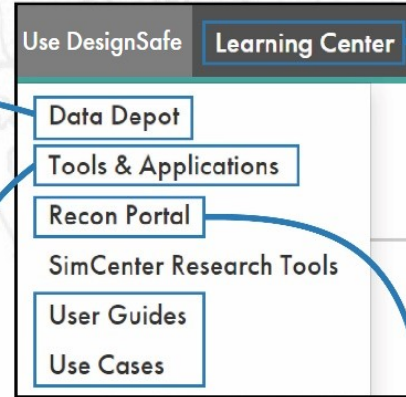
DesignSafe Components

www.designsafe-ci.org

Data



DATA DEPOT	
Find in My Projects	
Project ID	Title
PRJ-4836	Runout of Liquefaction-Induced Tailings Dam Failure: Sensitivity Analysis
PRJ-5626	Site amplification and nonlinearity
PRJ-4084	2023 NGA-Compatible CENA Database for GMM Validation and Site Response
PRJ-4684	Hybrid Finite Element and Material Point Method
PRJ-4784	Analysis of Mochikoshi Dam Failure using a Hybrid FEM-MPM Approach
PRJ-5581	Research Experiences for Undergraduates (REU), NSF NHERI 2024: Graph Networks
PRJ-1305	Community Data
PRJ-4767	Natural Hazards Research Summit 2024: Cyberinfrastructure for Natural Hazards

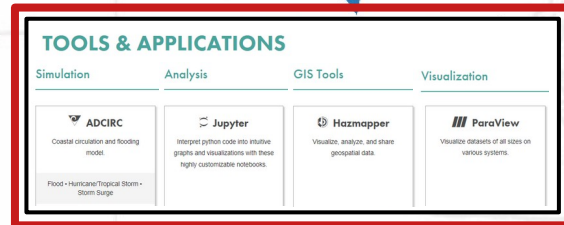


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Featured Playlists

- DesignSafe Webinars
- SimCenter Series: Studying Coastal Hazards with HydroUQ
- 2021 Joint NSF NHERI WOW and Lehigh RTMD EF User Workshop
- SimCenter Series: Advances in Computational Modeling and Simulation

Simulation & Data Analytics



Data



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Electronic Publishing of Research Products

DATA DEPOT

+ Add

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Published

Published (NEES)

Community Data

Help ▾

PRJ-3028 | Simulations of Earthquake-Induced Permanent Displacements of Generic Earth Slopes using LS-Dyna

Download Dataset

Hazard Type All Types ▾

Year Published

PI Rathje, Ellen

Project Type Simulation

Natural Hazard Type Earthquake

Keywords Seismic Slope Displacement, Predictive Modeling

[View Data Diagram](#) | [View Project Metrics](#) | [Leave Feedback](#)

Description | This DesignSafe published dataset is associated with research on seismic slope displacements through classical regression and artificial neural network (ANN) models. The database associated with different slope geometries, soil properties, and ground motion parameters for 1,051 earthquake motions. The maximum resultant displacements on the slope surface are presented.

[Show More](#)

Simulation | Slope Models in Finite Element Mesh Subjected to Numerical Earthquake Motions

Cite this Data:

Cho, Y., E. Rathje. (2021) "Slope Models in Finite Element Mesh Subjected to Numerical Earthquake Motions." *Displacements of Generic Earth Slopes using LS-Dyna*. DesignSafe Center for Data-Driven Engineering and Sensing.

Download Citation: [DataCite XML](#) | [RIS](#) | [BibTeX](#)

51 Downloads 593 Views 1 Citations [Details](#)

The uncertainties in the seismic slope displacement from ground motion characteristics and slope properties were investigated. It was conducted by running massive finite element simulations in parallel through HPC resources supported by Designsafe. Predictive models of the seismic slope displacement were derived from the computed displacement along with the ground motion intensity measure and slope parameters; Classical regression model and Artificial Neural Network regression model. These results could be reused for investigating fragility curve for slope failure.

Report | Description on dataset and method of simulations

Simulation Model | LS-Dyna Simulations for Seismic Slope Displacements

Simulation Input | Slope Models and Earthquake Records

☒ Name

Size

☐ Earthquake Motions (Velocity Time Series)

--

Boundary Condition

☐ LS-DYNA key files of all slope models

--

Simulation Script

Simulation Output | Compiled Output from Simulations

Analysis | Predictive Models of Seismic Slope Displacement



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Published

Published (NEES)

Community Data

Help ▾

PRJ-2998 | Machine Learning Models for the the Avon River Area Following the 2011 Christchurch Earthquake

Cite this Data:

Durante, M., E. Rathje. (2021) "Machine Learning Models for the the Avon River Area Following the 2011 Christchurch Earthquake." DesignSafe-CL. <https://doi.org/10.17862/10.17862/10.17862>

Download Citation: [DataCite XML](#) | [RIS](#) | [BibTeX](#)

142 Downloads 241 Views [Details](#)

Author(s) **Durante, Maria Giovanna; Rathje, Ellen**

Data Type **Jupyter Notebook**

Natural Hazard Type **Earthquake**

Date of Publication **03-12-2021**

[View Data Diagram](#) | [View Project Metrics](#) | [Leave Feedback](#)

Description | This project contains a Jupyter Notebook and python code for simulating lateral spreading displacement. This study is trained using data from the range of applicability reported in the ReadMe file. The Jupyter Notebook contains the ReadMe file.

[Simulations of Earthquake-Induced Lateral Spreading](#)

[Generic Earth Slopes using LS-Dyna](#)

[Simulation](#)

jupyter
nbviewer

JUPYTER

FAQ

</>

≡

↓

```
In [12]: #define function to find best parameter combination based on Cohen's kappa co
def rfr_model(X, y):
    # Perform Grid-Search
    kappa_scorer = make_scorer(cohen_kappa_score)
    gsc = GridSearchCV(
        estimator=RandomForestClassifier(),
        param_grid={
            'max_depth': range(2,10),
            'n_estimators': (5,10, 50, 100, 1000),
            'max_features': ('auto','sqrt','log2'),
            'criterion': ('gini','entropy'),
        },
        cv=10, verbose=0, n_jobs=-1,scoring=kappa_scorer)

    grid_result = gsc.fit(X, y)
    best_params = grid_result.best_params_

    rfr = RandomForestClassifier(max_depth=best_params["max_depth"],
                                n_estimators=best_params["n_estimators"],
                                max_features=best_params["max_features"], cr

    # Perform K-Fold CV
    scores = cross_val_score(rfr, X, y, cv=10)
    predictions = cross_val_predict(rfr, X, y, cv=10)
    optimised_random_forest = gsc.best_estimator_
```



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Jupyter Resources

JUPYTER OVERVIEW

The Jupyter Notebook is a web application that allows you to create and share documents that contain live code, equations, visualizations and explanatory text. Uses include: data cleaning and transformation, numerical simulation, statistical modeling, machine learning and much more.

Learn How to Use Jupyter

Application	Pro
Jupyter	No wait; No maximum job runtime limit; Sufficient memory for data processing purpose
Jupyter Lab HPC (CPU)	More memory and faster than Jupyter ; No need to reinstall Python packages once they have been installed; Excellent capability for data processing and parallel computing using 56 cores.
Jupyter Lab HPC (GPU)	Four GPUs in total, best performance for AI/ML jobs; Many commonly used Python packages for AI/ML applications have been pre-installed, including TensorFlow and PyTorch

Select a Version

Jupyter Hub

VM

Get Started

Access the Jupyter ecosystem.

Jupyter Lab HPC (CPU)

CPU

Get Started

Launch an interactive Jupyter instance running on a Frontera compute node. Community Data, Publications, and your HPC Work directory will be available within the session.

Jupyter Lab HPC (GPU)

GPU

Get Started

Launch an interactive Jupyter instance running on a Frontera RTX node with CUDA enabled. Community Data, Publications, and your HPC Work directory will be available within the session.



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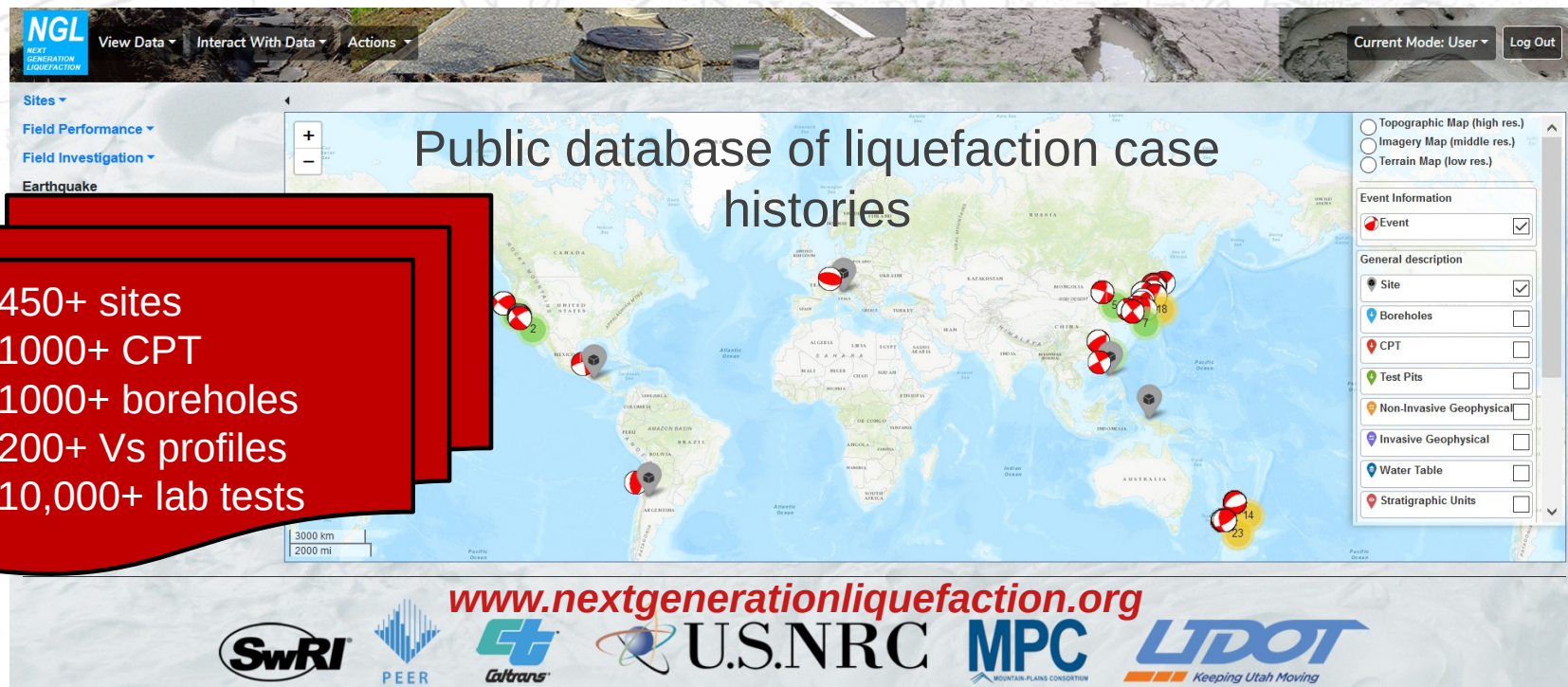
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Next Generation Liquefaction (NGL)



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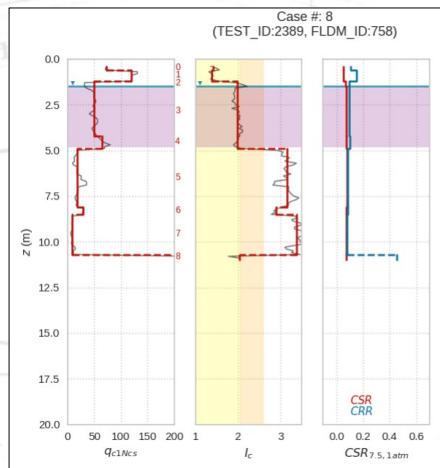
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RICE

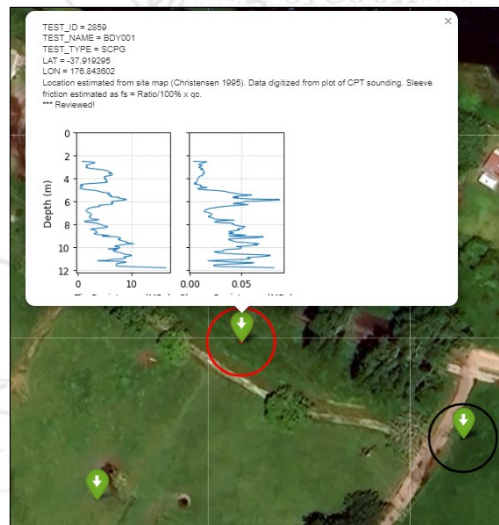
Florida Tech

DesignSafe access to NGL data

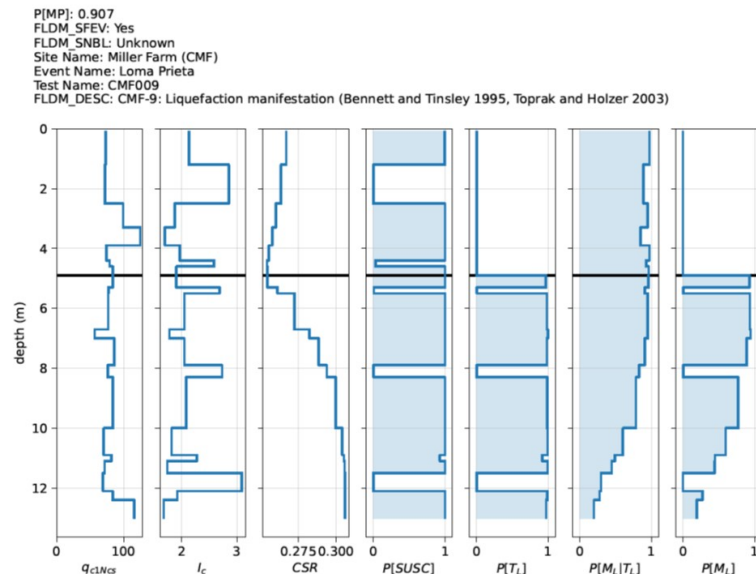
**Critical Layer
Evaluation**



Data Integration



**Bayesian liquefaction
model derived with pyMC**



From Kristin Ulmer (SWRI), Scott Brandenburg (UCLA)

<https://www.designsafe-ci.org/user-guide/usecases/brandenberg-ngl/usecase/>



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Final Thoughts

- AI / ML is being used across geotechnical engineering
 - Classification, regression, and so much more
 - Need a general understanding of methods, advantages/disadvantages
 - Beware of overfitting, “learning bad trends”, etc.
- Balance explainability with accuracy (XAI)



Final Thoughts

AI won't replace the scientist, but scientists who use AI will replace those who don't - David Womble, ORNL



The University of Texas at Austin

Maseeh Department of
Civil, Architectural and
Environmental Engineering
Cockrell School of Engineering



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Webinars and
tutorials



Prof. Ellen Rathje



Prof. Krishna Kumar



Geotechnical Research Group



TEXAS
The University of Texas at Austin

Online Continuing Education
Course on AI / ML

Acknowledgements

- Prof. Krishna Kumar (Univ of Texas)
- Dr. Margo Durante (Univ Calabria, Italy)
- Cheng-Hsi Hsiao (Univ of Texas)
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- Funding agencies and partners

